

Multicalibration for Unbiased Model-Based Prevalence Estimation

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Abstract

Political scientists increasingly use classifiers, and now large language models (LLMs), to estimate how prevalent a concept is in a corpus, such as topics in legislative agendas or categories in open-ended survey responses. Standard validation reports discriminative metrics such as accuracy or the area under the ROC curve (AUC) on a labeled sample. This certifies neither valid prevalence estimates in the validated population nor transportability to new ones. Because discrimination depends only on how a model ranks documents, it is blind to the feature-conditional errors that bias a prevalence estimate. Existing corrections either assume label shift or must be re-estimated for every target. We show that *multicalibration*, calibration conditional on input features rather than on average, is sufficient for unbiased prevalence estimation under any covariate shift along the calibrated features, and needs to be fitted only once. We demonstrate this in a simulation and in two applications, an LLM coding policy topics across four countries and languages, and a logistic regression predicting employment from survey data. Multicalibration holds near-zero bias under shift.

Introduction

Machine learning classifiers, and increasingly large language models (LLMs), have become standard tools to estimate the prevalence of a concept in political text corpora (Baumgartner et al. 2006; Benoit et al. 2026; Overos et al. 2024; King and Lowe 2003; Weidmann et al. 2026; Mellon et al. 2024; Gilardi et al. 2023; Ornstein et al. 2025). The quantity of interest is usually not the label on any single document but an *aggregate* count or prevalence (González et al. 2017; Hopkins and King 2010).

The standard way to validate such a classifier is to measure how well it discriminates: how closely its labels agree with human codes, reported as accuracy, F_1 , or the area under the ROC curve (AUC) and judged against the intercoder reliability of the human benchmark (Halterman and Keith 2025; Hayes and Krippendorff 2007). For prevalence estimation this is not sufficient, for two related reasons. First, discrimination does not imply an unbiased prevalence estimate even on the population the classifier was validated on. A classifier can rank documents almost perfectly yet misstate the proportion that are positive, because its false positives and false negatives need not cancel. The quantification literature corrects this, adjusting the raw count with error rates estimated on the labeled sample (González et al. 2017).

Second, that correction is tied to the population it was estimated on. When the same instrument is applied to a new target, for example to compare sub-corpora or track a quantity over time, it no longer holds in general. This is due to covariate shift. Across targets, the mix of documents changes (languages, venues, eras, authors) while the rule linking content to category stays fixed (e.g., through the codebook). When a target holds more of the cases the classifier misjudges, errors that canceled on the validation sample no longer do. Classical quantifiers assume the opposite. Under label shift, documents of a category look the same everywhere and only their overall frequency changes, which rarely holds across languages and venues. A high AUC does not protect against this, and standard validation will not reveal it (Adcock and Collier 2001).

Existing tools each address part of the problem, but none yields a *reusable* instrument. Quantification methods estimate prevalence from imperfect classifiers (González et al. 2017), but Classify & Count, its bias-corrected relatives (Rogan and Gladen 1978; Saerens et al. 2002; Maletzke et al. 2018), and the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023) all assume label shift, and so do not transport under covariate shift. Design-based supervised learning (Egami et al. 2023), prediction-powered inference (Angelopoulos et al. 2023), and importance weighting (Shimodaira 2000; Sugiyama et al. 2007) apply under covariate shift but must be re-estimated for each target. A related literature on learned proxies (Knox et al. 2022) and the reliability of LLM annotations (Bisbee et al. 2024; Argyle et al. 2023; Rodriguez and Spirling 2022) raises the same worry that imperfect measures distort inference; Weidmann et al. (2026) find LLM classifiers deviate from experts in systematic, feature-dependent directions that conditioning on features can correct.

The simplest model-based prevalence estimator just averages the model’s outputs (scores or labels). Such an estimator is unbiased only if the classifier is *calibrated*, that is, if its outputs can be interpreted as predicted probabilities that match empirical frequencies in the population at hand. Calibration, however, is a property of one population and need not survive a shift. Empirically, it degrades under covariate shift even when accuracy is preserved (Yang et al. 2023; Ren et al. 2025). Confidence elicitation (Tian et al. 2023; Kadavath et al. 2022; Wang et al. 2023) and post-hoc recalibration (Wu and Resnick 2024) aim only at this global calibration. We show that *multicalibration*, calibration conditional on the input features rather than on average (Hébert-Johnson et al. 2018; Detommaso et al. 2024), fills this gap. A multicalibrated classifier is stable under reweighting of those features. Calibrated once on a labeled source sample, it returns unbiased prevalence estimates on any target whose shift the features capture and that lies within their support (Kim et al. 2022).

Prevalence Bias Reduction through Multicalibration

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a predictor $h(X) \in [0, 1]$. The estimand is the prevalence $\pi^* = P^*(Y = 1)$ in a target population, estimated from unlabeled target data $\{X_i\}$ and h . We assume *covariate shift*, which means that $P(X)$ differs between calibration and target while $P(Y | X)$ is stable. Covariate shift is the complement of the *label-shift* assumption, which instead holds $P(X | Y)$ fixed while the prior $P(Y)$ moves.¹ Under concept drift, where $P(Y | X)$ itself changes across contexts, no purely statistical method recovers the target prevalence without labels from the target.

A predictor is *globally calibrated* if $\mathbb{E}[Y | h(X) = p] = p$ for all p . Then $\mathbb{E}[h(X)] = \pi$, and averaging

¹The causal heuristic behind these invariance assumptions traces to @scholkopf2012causal; we rely on the invariance, not the causal direction.

its predictions is unbiased. But calibration on one population need not hold on another. Partition the population into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y \mid X \in G]$ be the predictor’s mean signed error in G . Global calibration requires only $\sum_G w_G \epsilon_G = 0$. Under covariate shift the weights become w_G^* , and the prevalence bias is

$$\hat{\pi}^* - \pi^* = \sum_G w_G^* \epsilon_G,$$

which is nonzero in general unless every $\epsilon_G = 0$.² A predictor can be perfectly calibrated on the validation sample yet badly biased on a target that reweights those groups. High classification accuracy or discriminative performance does not guard against this problem. AUC and related discriminative measures depend only on how the predictor ranks documents, so they are invariant to any strictly increasing transformation of the scores. Such a transformation leaves every discriminative metric untouched while moving each ϵ_G , and with it the prevalence estimate, essentially arbitrarily. Accuracy and F_1 are not rank-invariant, but they count errors without their sign, whereas in a prevalence estimate a false positive and a false negative cancel. Accuracy is also itself an average over the same groups, so a classifier can be highly accurate overall and still make large, same-signed errors in a group that is small on the source and large on the target.

Existing methods for correction of this bias fail because none removes the group errors ϵ_G themselves. Output-based corrections (Classify & Count, Rogan-Gladen, global isotonic recalibration (Wu and Resnick 2024)) estimate aggregate positive and error rates on the source; these are weighted averages over groups, and they no longer cancel once the target reweights the groups. Label-shift quantifiers (Saerens-Latinne-Decaestecker, SLD, also known as EMQ (Saerens et al. 2002), distribution-matching, the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023)) fail for a different reason. They carry no per-document ϵ_G , and the identity they solve presumes $P(X \mid Y)$ is stable, which covariate shift violates by construction. SI Section S7 gives the argument and benchmarks the canonical political-science case (ReadMe) directly.

Multicalibration drives every ϵ_G to zero, not merely their weighted sum, by replacing h with a predictor f calibrated within each group. Formally, f is multicalibrated with respect to a class of groups $\mathcal{G}$ if $\mathbb{E}[Y \mid f(X) = v, X \in G] = v$ for all $G \in \mathcal{G}$ and values v (Hébert-Johnson et al. 2018).³ A multicalibrated predictor then satisfies the Kim et al. (2022) “universal adaptability” guarantee. For any target whose density ratio dP^*/dP is measurable with respect to $\mathcal{G}$,

$$\mathbb{E}^*[f(X)] = \mathbb{E}\left[\frac{dP^*}{dP}(X) f(X)\right] = \mathbb{E}\left[\frac{dP^*}{dP}(X) Y\right] = \mathbb{E}^*[Y] = \pi^*,$$

because the density ratio is constant within each $\mathcal{G}$ -cell, so the within-cell $\epsilon_G = 0$ carries through. Therefore, the multicalibrated instrument provides an unbiased prevalence estimate on every target whose shift lies along the calibrated features and within support, with no knowledge of the shift and no target-specific estimation.

This bias reduction rests on two conditions. First, the target’s feature values must lie within the calibration *support*, which can be tested from unlabeled target data. Second, the feature class $\mathcal{G}$ must be *expressive* enough to capture every dimension along which the shift interacts with the outcome. Expressiveness is an ignorability condition, which is not verifiable from data. It rests

²Exact when the density ratio is constant within each group, a first-order approximation otherwise; see SI Section S1.

³The weaker *multi-accuracy* (zero average error per group) already suffices for unbiased prevalence; see SI Section S1.8.

on the same substantive judgment as any observational adjustment. So does the covariate-shift assumption itself. Whether a shift is covariate rather than label shift or concept drift cannot be read off unlabeled data and must be argued from how the concept is defined. However, since a multicalibrated predictor is also globally calibrated, and enlarging $\mathcal{G}$ tends to reduce the residual bias, a large number of features can be included in the multicalibration model. With a properly regularized multicalibration model, there is almost no downside to including such a large number of features and the resulting multicalibrated model usually strictly improves over the base model (see Tax et al. (2026) for some empirical evidence of this claim). We obtain the multicalibrated predictor with MCGrad (Tax et al. 2026), a post-hoc gradient-boosting multicalibrator.

Simulation

We first illustrate the mechanism in a setting simple enough to verify by hand. We generate synthetic data with a binary covariate $X \in \{0, 1\}$ and outcome Y with $P(Y = 1 | X = 1) = 0.85$ and $P(Y = 1 | X = 0) = 0.15$. A classifier produces deterministic, systematically biased scores (a 10% multiplicative bias within each stratum, so $\hat{p} = 0.135$ when $X = 0$ and $\hat{p} = 0.935$ when $X = 1$). We then estimate prevalence on target distributions with $P(X = 0)$ ranging from 0.01 to 0.99, holding all calibration parameters fixed at their training values, over 50 replications.

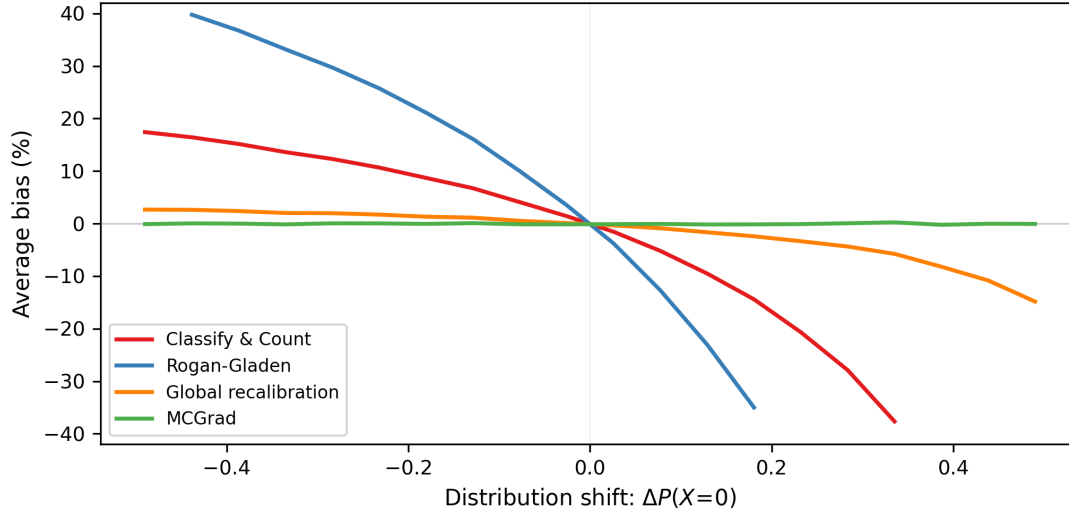


Figure 1. Relative prevalence bias under covariate shift (50 runs). The x-axis is the change in $P(X = 0)$ from the training value of 0.5. Classify & Count and Rogan-Gladen diverge with shift; global recalibration shows moderate bias; the multicalibrated estimator stays near zero. Cropped at $\pm 40\%$; the full range, additional methods, and RMSE are in SI Sections S4–S5.

At the training distribution ($\Delta P(X = 0) = 0$) every method is approximately unbiased (Figure 1). As the target shifts, they diverge. Classify & Count and Rogan-Gladen grow with the shift, the latter’s ratio form amplifying the error beyond $\pm 40\%$. Global recalibration shows moderate but nonzero bias, up to roughly 15% at the most extreme shift. Only the multicalibrated estimator stays near zero throughout. The estimators in Figure 1 differ only in how it turns the classifier’s scores into a prevalence; all share the same ranking of X , and so have identical discriminative performance.

Applications

We demonstrate the mechanism on two applications with known ground truth. The first is an LLM coding political text. We use Claude Opus 4.6⁴ to classify 30,000 Comparative Agendas Project texts (Baumgartner et al. 2006) for whether they concern Law & Crime (CAP major topic 12), scored against the published expert codes. The documents span six sub-populations across four countries and languages, 5,000 each. Law & Crime is a rare topic, coded in 7.9% of documents at baseline and 6–20% across scenarios. The second is an ordinary logistic regression predicting employment from 16 sociodemographic features in American Community Survey (ACS) microdata. In both, we construct targets whose covariate shift is known by design: within-support reweightings (of countries and document types; of age groups) and out-of-support targets (unseen document types; held-out states). The LLM is prompted for both a Yes/No classification and an elicited probability $P(\text{Yes})$ (Tian et al. 2023; Kadavath et al. 2022). We multicalibrate on the bare Yes/No label, the output essentially all applied studies produce, and reserve the probability for the methods that require a score. Designs, feature sets, and full numbers are in SI Sections S2–S3.

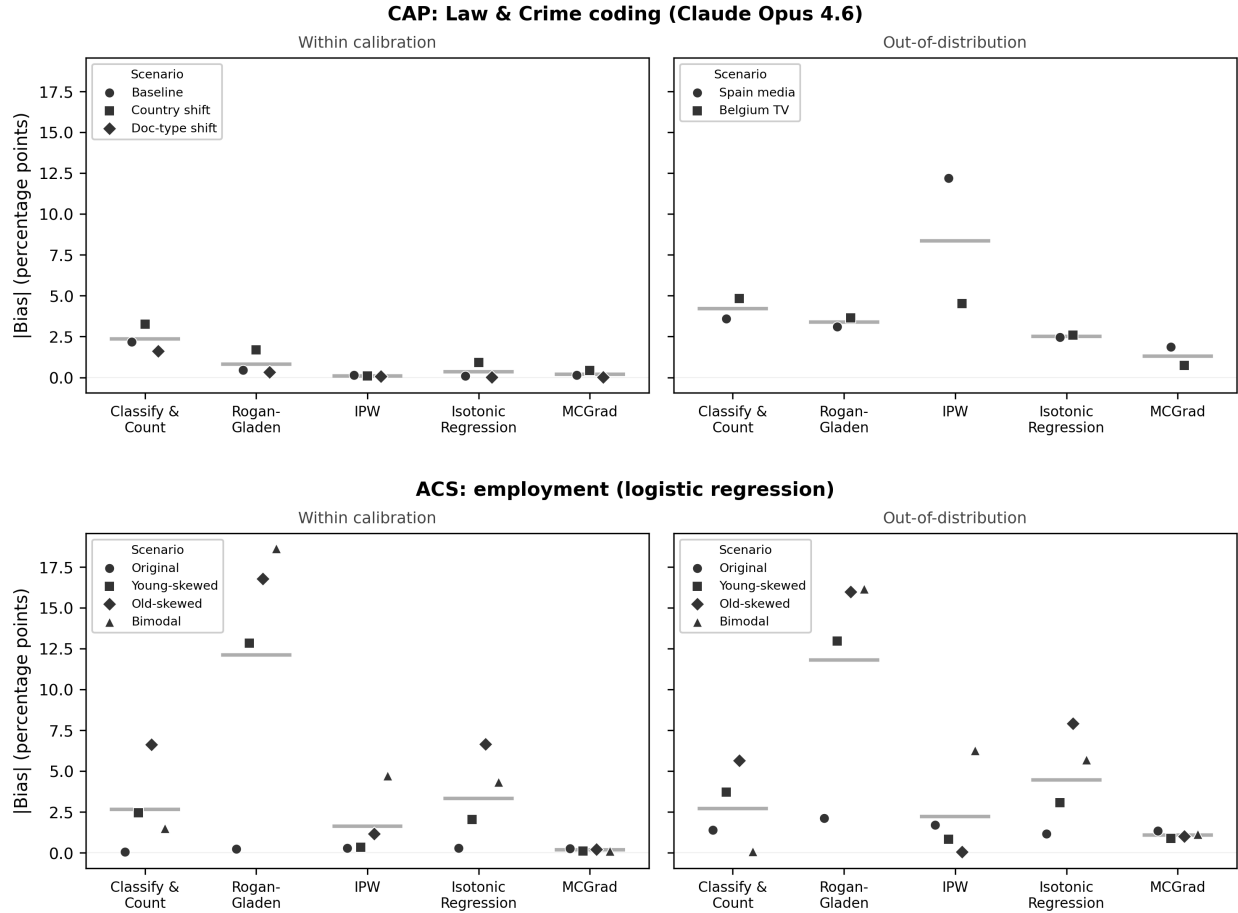


Figure 2. Absolute prevalence bias (percentage points) by method and scenario. Top: CAP Law & Crime coding with Claude Opus 4.6, multicalibrated on the Yes/No label. Bottom: ACS employment with logistic regression. Left: targets within the calibration support; right: out-of-distribution targets. Marker shape denotes scenario, horizontal bars are scenario means, and all panels share one scale.

⁴An open-weights replication with Llama 3.3 70B (SI Section S2) confirms the findings and permits full reproduction.

Full numbers in SI Tables S1–S2.

In the political-text application the LLM clears any discriminative-validation bar, with per-language AUC of 0.98–0.99, yet Classify & Count, the applied default, is biased by +1.6 to +4.8pp (Figure 2, top), a relative overstatement of nearly 30% at baseline and over 40% at its worst. Rogan-Gladen reduces this within support but still misses by +3.1 to +3.7pp out of support. SLD, the label-shift quantifier, is already biased at baseline (+7.4pp, since the elicited scores are not the calibrated posteriors its EM step assumes) and reaches +21.6pp out of support, confirming that the label-shift family does not transport under covariate shift.⁵ Isotonic recalibration of the scores is accurate within support (≤ 0.9 pp) but drifts out of it (+2.6pp on Belgian TV, -2.5 pp on Spanish media). Inverse probability weighting (IPW) is near-zero within support and fails out of it (-12.2 pp), where the target contains feature values absent from the source (a positivity violation). Multicalibration on the bare Yes/No labels achieves near-zero bias within support (≤ 0.4 pp) and degrades gracefully out of support (+0.7pp on Belgian TV, -1.9 pp on Spanish media). Feeding it the elicited probabilities instead performs comparably within support and somewhat worse out of it (SI Table S2), because the correction comes from the metadata features and the calibration sample, not from the quality of the model’s confidences.

The survey application reproduces this ordering with a wider spread (Figure 2, bottom). Rogan-Gladen and SLD fail by 12–19pp under the age shifts. Classify & Count and isotonic regression stay closer but grow with the shift, reaching 6–8pp at the largest ones. IPW is accurate on the simple age shifts (≤ 1.2 pp) but breaks on the bimodal one (+4.7pp), where the density ratio is hard to model. Multicalibration stays at ≤ 0.27 pp across every within-support target, including the bimodal shift, and at 0.88–1.35pp on held-out states. Read together, the two applications locate the condition under which global recalibration is sufficient: it works when the predictor’s errors are roughly homogeneous across the groups a shift reweights, which the CAP shifts approximately satisfy. The ACS shifts move age, which drives employment hard (76% employed at ages 25–54 against 17% at 65+), so group errors are large and heterogeneous, and only feature-conditional calibration removes them.

Discussion

Prevalence estimation across populations is a common goal in political science research. With LLMs becoming more popular, even more zero-shot, model-based measurement of prevalence is being unlocked. Discriminative validation shows only that a classifier separates classes. It is silent on whether the resulting prevalence estimates are unbiased, either in the population where discrimination was validated or in any other. In our LLM application a classifier with per-language AUC above 0.98 still produced default prevalence estimates off by roughly 30–40% of the true rate, even without shift and worse under it, and would have passed any discriminative check while misreporting prevalence. Multicalibration, calibration conditional on features rather than on average, removes this bias where discrimination was validated and keeps it removed as the target shifts.

However, the guarantee has limits. It holds only along calibrated dimensions, so calibration data must span the anticipated variation. It also requires labeled calibration data, which in zero-shot settings reintroduces some hand-coding, though far less than per-document labeling and only once.

⁵ReadMe [@hopkinsking2010nonparametric], the canonical political-science quantifier, is classifier-free and fails as the label-shift diagnosis predicts, exceeding +60pp under some shifts; details and ReadMe2 [@jerkzackingstrezhev2023improved] in SI Section S7 (Table S4).

Under mixtures the covariate-shift condition holds only approximately. Our applications have ground truth, so we can verify the estimates directly; a practitioner usually cannot, since the target’s labels are what is missing.

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